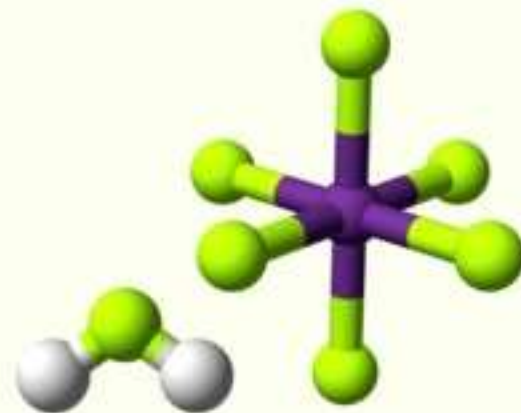


CHEMISTRY

THE PERIODIC TABLE

MODULE - 6



LOOK AT THE FOLLOWING



A HEAP OF CLOTHES

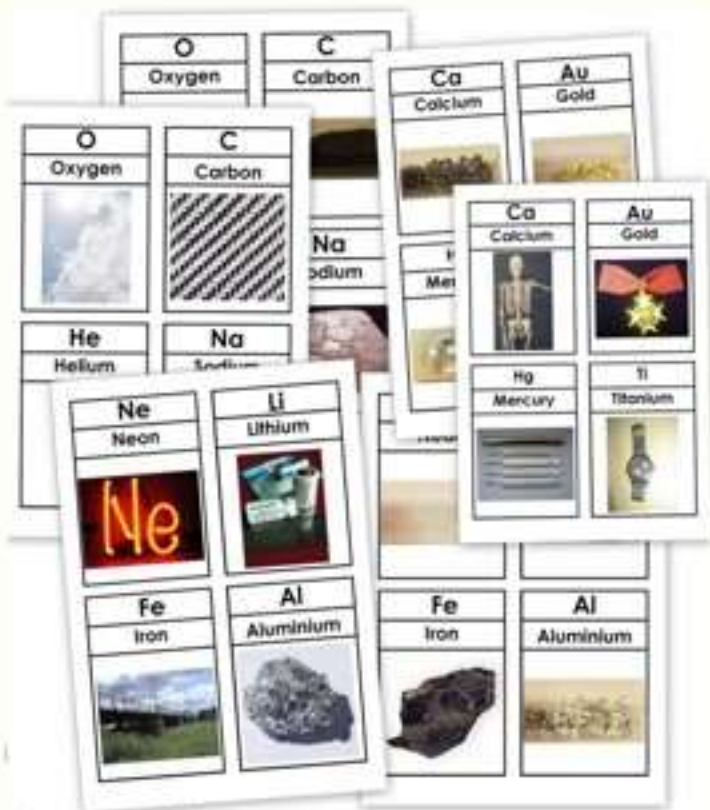


A SHOPPING MALL

WHICH IS THE RIGHT PLACE TO FIND A DRESS?



WHAT IS THE NEED FOR CLASSIFICATION?



- There are multiple elements in our Universe.
- To study the physical and chemical properties of the elements, they need to be classified.
- Classification based on some fundamental properties will help to group similar elements and to study them easily.
- Thus, a need for classification of elements arose.



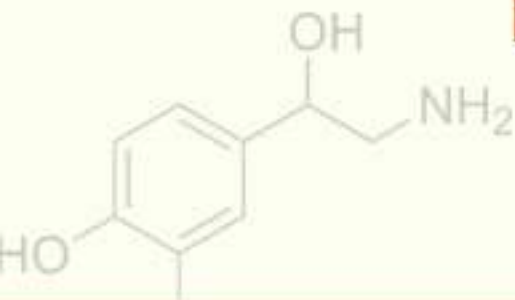
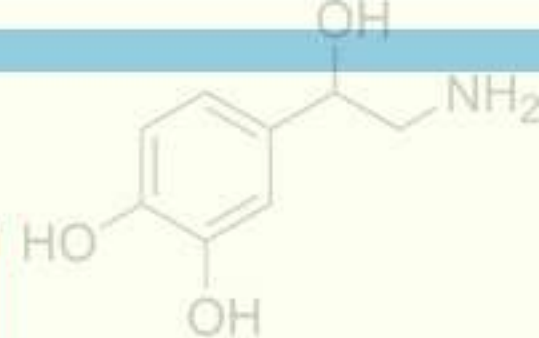
THE CLASSIFICATION OF ELEMENTS



DOBEREINER

LAW OF TRIADS

- Some groups of elements show similar properties. These are called as the Triads.
- In the three elements, the middle element has atomic mass which is approximately the mean of the atomic mass of the other two elements.



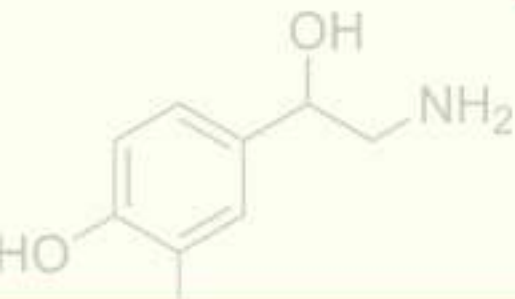
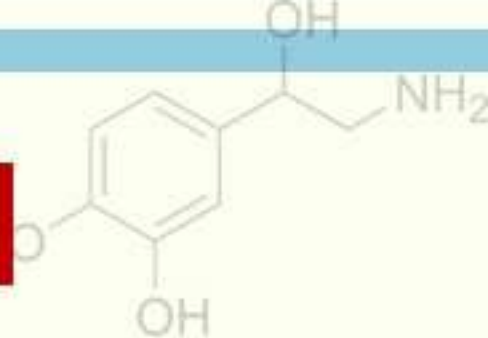
THE CLASSIFICATION OF ELEMENTS



NEWLANDS

LAW OF OCTAVES

- Of the 56 elements were added in increasing order of their atomic masses.
- Every eighth element had similarity to the first element.
- It was comparable with the musical notes of octaves.



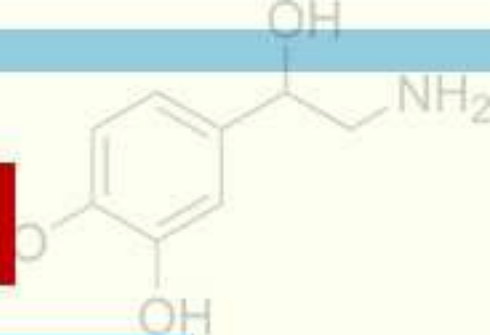
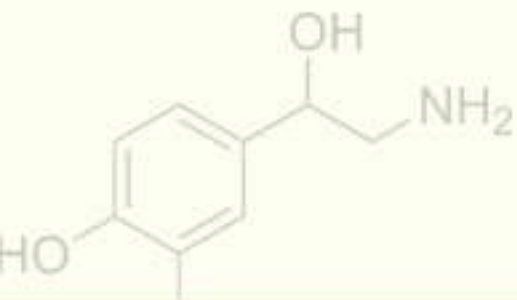
THE CLASSIFICATION OF ELEMENTS



MENDELEEV

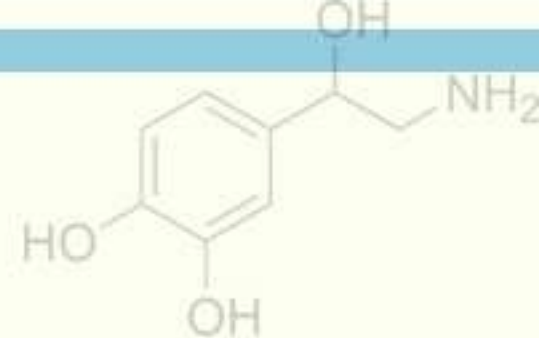
MENDELEEV'S PERIODIC TABLE

- Mendeleev arranged the then known 63 elements in increasing order of atomic mass.
- He classified the elements with similar physical and chemical properties for the 1st time.
- **Mendeleev's Periodic law stated that the properties of elements are periodic functions of their atomic masses.**



THE CLASSIFICATION OF ELEMENTS

MENDELEEV'S PERIODIC TABLE



O	I	II	III	IV	V	VI	VII	VIII			
He 4.00	H 1.01	Li 6.94	Be 9.01	B 10.8	C 12.0	N 14.0	O 16.0	F 19.0			
Ne 20.2	Na 23.0	Mg 24.3	Al 27.0	Si 28.1	P 31.0	S 32.1	Cl 35.5				
Ar 40.0	K 39.1	Ca 40.1	Sc 45.0	Ti 47.9	V 50.9	Cr 52.0	Mn 54.9	Fe 55.9	Co 58.9	Ni 58.7	
	Cu 63.5	Zn 65.4		Ga 69.7	Ge 72.6	As 74.9	Se 79.0				Br 79.9
Kr 83.8	Rb 85.5	Sr 87.6	Y 88.9	Zr 91.2	Nb 92.9	Mo 95.9	Tc (99)	Ru 101	Rh 103	Pd 106	
	Ag 108	Cd 112		In 115	Sn 119	Sb 122	Te 128				I 127
Xe 131	Ce 133	Ba 137	La 139	Hf 179	Ta 181	W 184	Re 180	Os 194	Ir 192	Pt 195	
	Au 197	Hg 201		Tl 204	Pb 207	Bi 209	Po (210)				At 210
Rn 222	Fr (223)	Ra (226)	Ac (227)	Th 232	Pa (231)	U 238					

 Dobereiner's triads

 known to Mendeleev

 lanthanides

 actinides

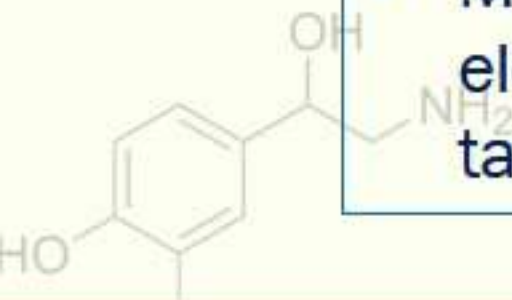
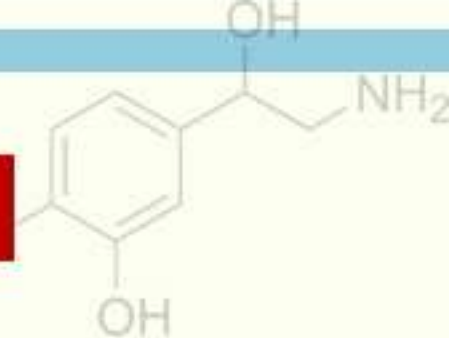
 known to the ancients



WHY MENDELEEV'S PERIODIC TABLE IS IMPORTANT?

SOME KEY POINTS

- The major credit for a systematic classification of elements goes to Mendeleev.
- Mendeleev was able to predict the existence of undiscovered elements like Ga, Sc and Ge
- He named them eka- boron(Scandium), eka- alumunium (Gallium), eka-silicon(Germanium) and also correctly predicted their properties.
- Mendeleev corrected the atomic masses of a few elements on the basis of their positions in the periodic table to keep same elements together.



WHY DID MENDELEEV'S PERIODIC TABLE FAILED ?

- Position of Hydrogen was not proper in his Table, as hydrogen shows both the properties of a metal and a non-metal.
- Isotopes have not been given separate places in the periodic table because of their similar properties, but they do differ in their atomic masses!
- In certain pair of elements, the increasing order of atomic masses was not obeyed to keep the similar elements together.
- Some similar elements are separated, in the periodic table. For example copper (Cu) and mercury (Hg), barium (Ba) and lead (Pb). On the other hand, some dissimilar elements have been placed together in the same group.
- Mendeleev could not explain the cause of periodicity among the elements.



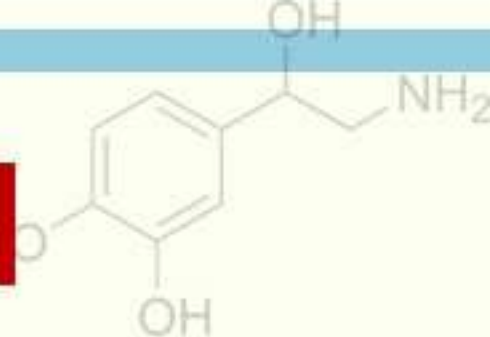
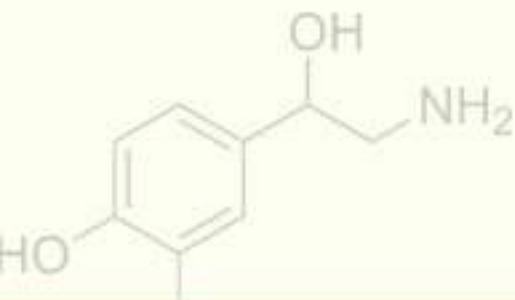
THE CLASSIFICATION OF ELEMENTS

THE MODERN PERIODIC TABLE



MOSELEY

- Moseley reconstructed the Periodic Table that was devised by Mendeleev.
- The elements of the Modern Periodic Table were arranged in increasing order of their atomic numbers.
- **The Modern Periodic law states that the properties of the elements are periodic functions of their atomic numbers.**



THE MODERN PERIODIC TABLE

- hydrogen
- alkali metals
- alkali earth metals
- transition metals
- poor metals
- nonmetals
- noble gases
- rare earth metals

1

H

2

He

3

Li

4

Be

5

B

6

C

7

N

8

O

9

F

10

Ne

11

Na

12

Mg

13

Al

14

Si

15

P

16

S

17

Cl

18

Ar

19

K

20

Ca

21

Sc

22

Ti

23

V

24

Cr

25

Mn

26

Fe

27

Co

28

Ni

29

Cu

30

Zn

31

Ga

32

Ge

33

As

34

Se

35

Br

36

Kr

37

Rb

38

Sr

39

Y

40

Zr

41

Nb

42

Mo

43

Tc

44

Ru

45

Rh

46

Pd

47

Ag

48

Cd

49

In

50

Sn

51

Sb

52

Te

53

I

54

Xe

55

Cs

56

Ba

57

La

72

Hf

73

Ta

74

W

75

Re

76

Os

77

Ir

78

Pt

79

Au

80

Hg

81

Tl

82

Pb

83

Bi

84

Po

85

At

86

Rn

87

Fr

88

Ra

89

Ac

104

Unq

105

Unp

106

Unh

107

Uns

108

Uno

109

Une

110

Unn

hydrogen

alkali metals

alkali earth metals

transition metals

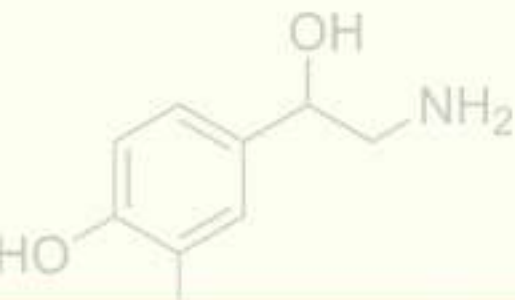
poor metals

nonmetals

noble gases

rare earth metals

⁵⁸ Ce	⁵⁹ Pr	⁶⁰ Nd	⁶¹ Pm	⁶² Sm	⁶³ Eu	⁶⁴ Gd	⁶⁵ Tb	⁶⁶ Dy	⁶⁷ Ho	⁶⁸ Er	⁶⁹ Tm	⁷⁰ Yb	⁷¹ Lu
⁹⁰ Th	⁹¹ Pa	⁹² U	⁹³ Np	⁹⁴ Pu	⁹⁵ Am	⁹⁶ Cm	⁹⁷ Bk	⁹⁸ Cf	⁹⁹ Es	¹⁰⁰ Fm	¹⁰¹ Md	¹⁰² No	¹⁰³ Lr

NC(CO)c1ccc(O)cc1

THE MODERN PERIODIC TABLE - BREAKUP

TOTAL NUMBER OF ELEMENTS

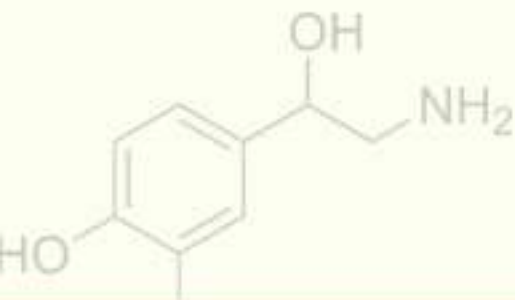
118

ELEMENTS FOUND IN NATURE

92

STABLE ELEMENTS

83



THE MODERN PERIODIC TABLE - BREAKUP

H ¹	
Li ³	He ²
Na ¹¹	Ne ¹⁰
K ¹⁹	Ar ¹⁸
Rb ³⁷	Kr ³⁶
Cs ⁵⁵	Xe ⁵⁴
Fr ⁸⁷	Rn ⁸⁶

GROUPS

- The vertical columns of the Periodic Table are called groups or families.
- There are 18 groups in the Modern periodic table.
- The elements in the groups exhibit similar properties.
- The Valency of the members of the group is same.



THE MODERN PERIODIC TABLE - BREAKUP

BLOCKS

S - block

p - block

d - block

f - block

s-Block

H	
Li	Be
Na	Mg
K	Ca
Rb	Sr
Cs	Ba
Fr	Ra

d-Block

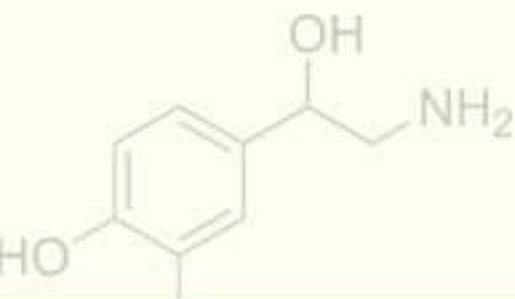
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Lr	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn

p-Block

B	C	N	O	F	Ne
Al	Si	P	S	Cl	Ar
Ga	Ge	As	Se	Br	Kr
In	Sn	Sb	Te	I	Xe
Tl	Pb	Bi	Po	At	Rn
Uut	Fl	Uup	Lv	Uus	Uuo

f-Block

La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No

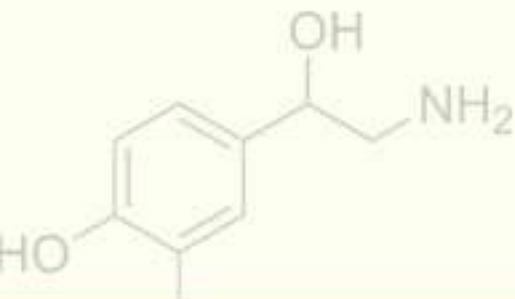


THE MODERN PERIODIC TABLE - BREAKUP

H	
Li	Be
Na	Mg
K	Ca
Rb	Sr
Cs	Ba
Fr	Ra

s - block

- Based on the orbital system.
- Has 2 groups
- 1st group is Alkali Metals
- 2nd group is Alkali Earth Metals



THE MODERN PERIODIC TABLE - BREAKUP

p - block

					He ²
B ⁵	C ⁶	N ⁷	O ⁸	F ⁹	Ne ¹⁰
Al ¹³	Si ¹⁴	P ¹⁵	S ¹⁶	Cl ¹⁷	Ar ¹⁸
Ga ³¹	Ge ³²	As ³³	Se ³⁴	Br ³⁵	Kr ³⁶
In ⁴⁹	Sn ⁵⁰	Sb ⁵¹	Te ⁵²	I ⁵³	Xe ⁵⁴
Tl ⁸¹	Pb ⁸²	Bi ⁸³	Po ⁸⁴	At ⁸⁵	Rn ⁸⁶

- The only group that consists of non- metals.
- Has 6 groups
- 3 group (*boron family*)
- 4 group (*carbon family*)
- 5 group (*nitrogen family*)
- 6 group (*oxygen family*)
- 7 group (*halogens*)
- 8 group (*noble gases*)

P block contains the metalloids that are just at the boundary of the zig zag line

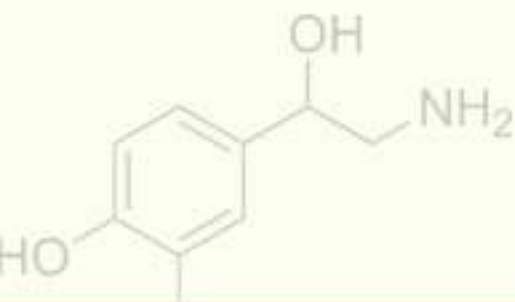


THE MODERN PERIODIC TABLE - BREAKUP

d - block

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Lr	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn

- D- block is also called the transition elements.
- The elements show some characteristic color, also have some catalytic effects.
- The d- block elements are para-magnetic in nature.

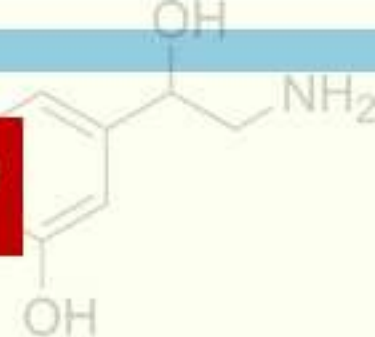
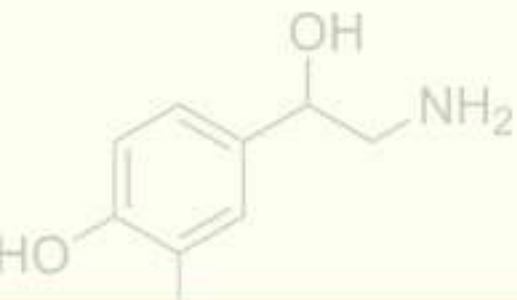


THE MODERN PERIODIC TABLE - BREAKUP

f - block

La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No

- They are called the inner Transition elements.
- They constitute of the rare earth metals
- The elements have high atomic mass and are mostly radioactive.
- They are written at the bottom of the Main Periodic Table because these group is curved out of d-block to keep the aesthetic of group intact.



THE MODERN PERIODIC TABLE - BREAKUP

PERIODS

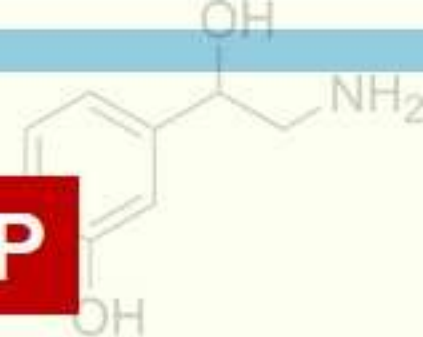
- The horizontal rows are the periods of the Periodic Table.
- There are 7 rows of the periodic table.
- The elements arranged in the rows show different properties.
- But, they all have the same number of shells.

11 Na	12 Mg	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
----------	----------	----------	----------	---------	---------	----------	----------

19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
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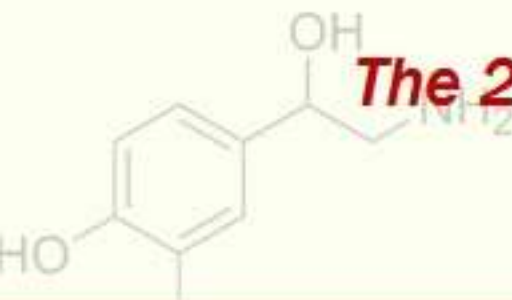


THE MODERN PERIODIC TABLE - BREAKUP



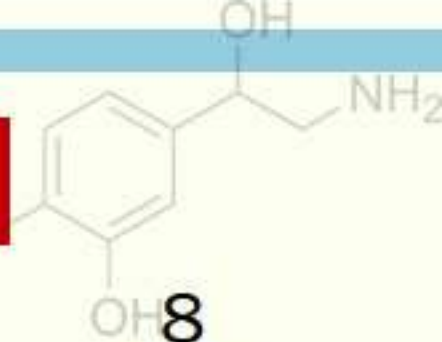
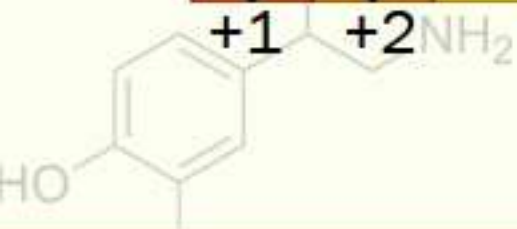
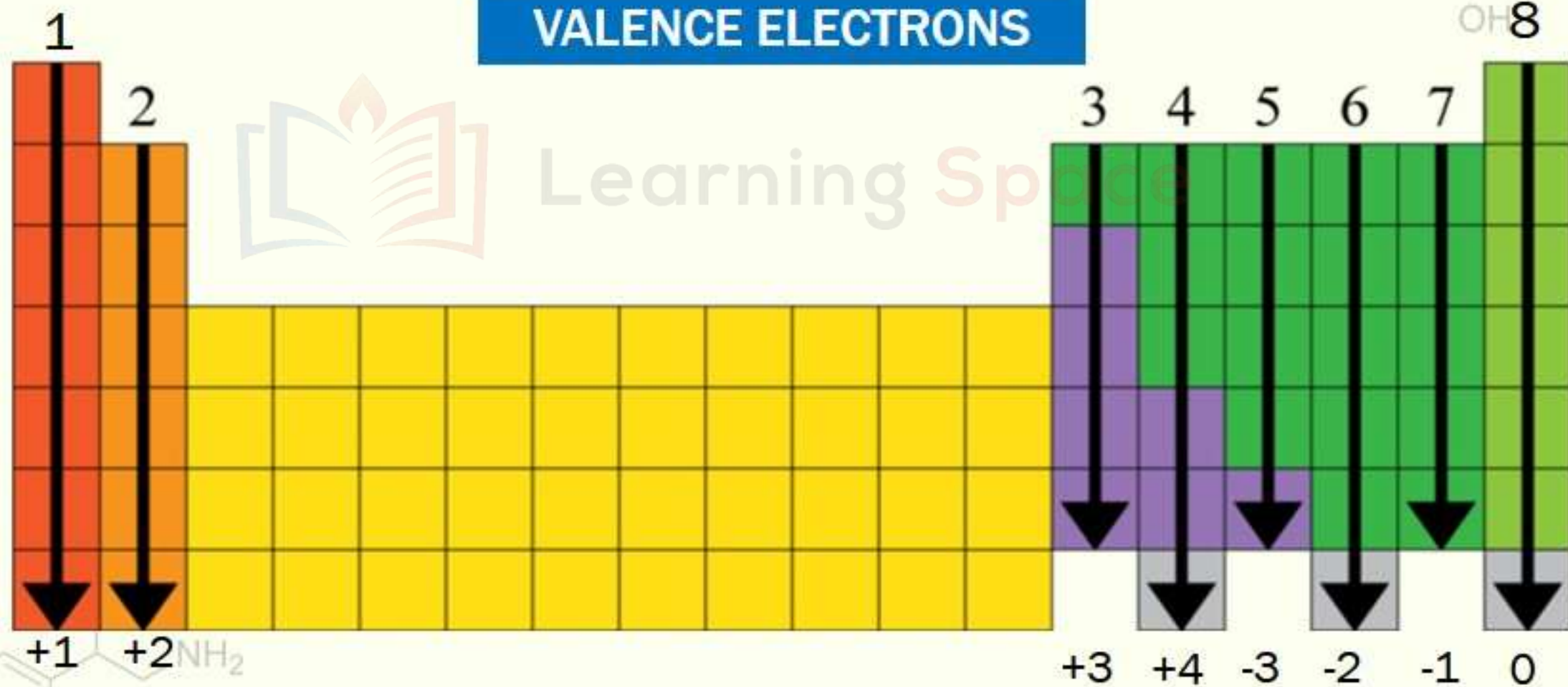
PERIOD	NO. OF ELEMENTS	NAME
1 st	2	Very Short Period
2 nd & 3 rd	8	Short Period
4 th & 5 th	18	Long Period
6 th	32 (lanthanides)	Very Long period
7 th	— (Actinides)	Incomplete

The 2nd Period is called Bridge elements and the 3rd Period is called Typical Elements.



TRENDS IN MODERN PERIODIC TABLE

VALENCE ELECTRONS



NC(CO)c1ccc(O)c(O)c1CC(O)C1=CC=C(C=C1)O

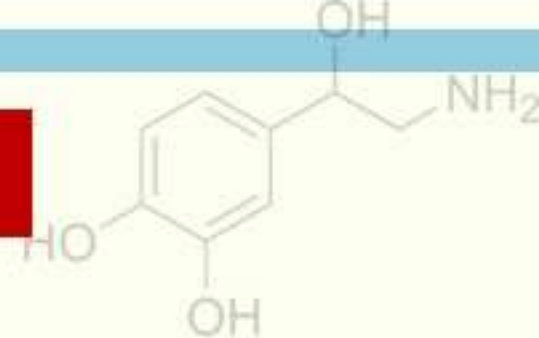
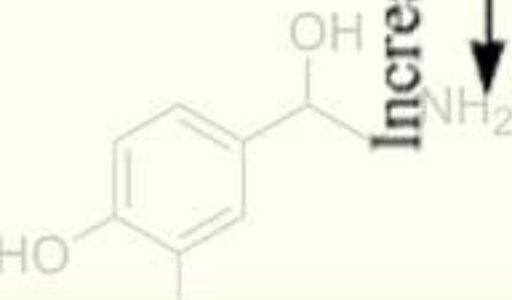
TRENDS IN MODERN PERIODIC TABLE

METALLIC CHARACTER

Increasing Metallic Character

Increasing metallic character

Metallic Character



TRENDS IN MODERN PERIODIC TABLE

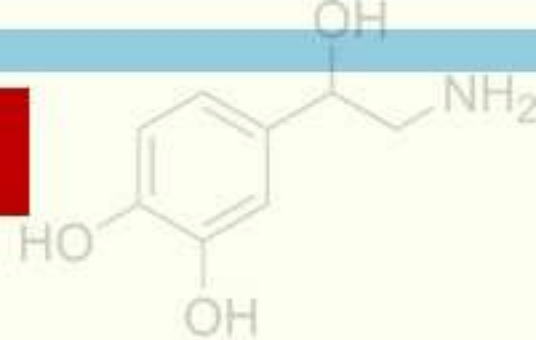
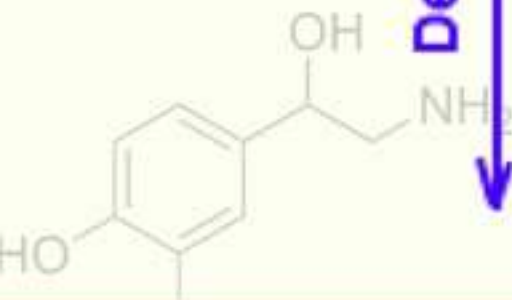
ELECTRONEGATIVITY

Increases

Decreases

Electronegativity

Decreases



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